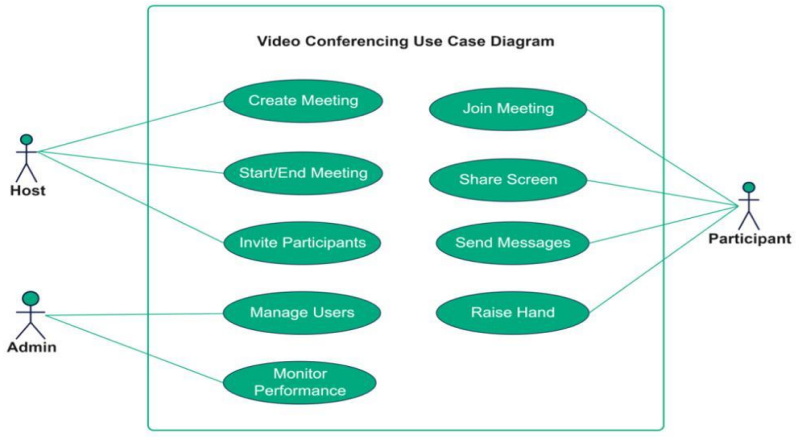
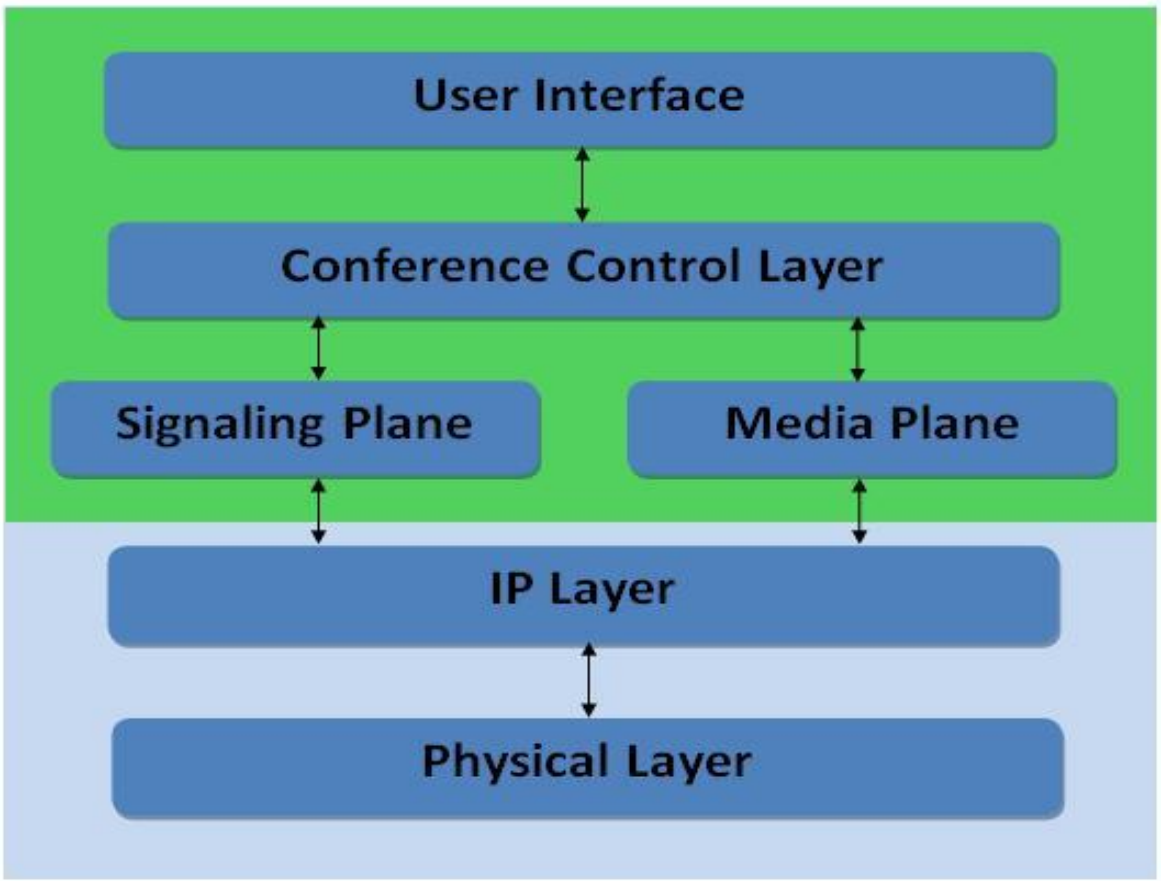
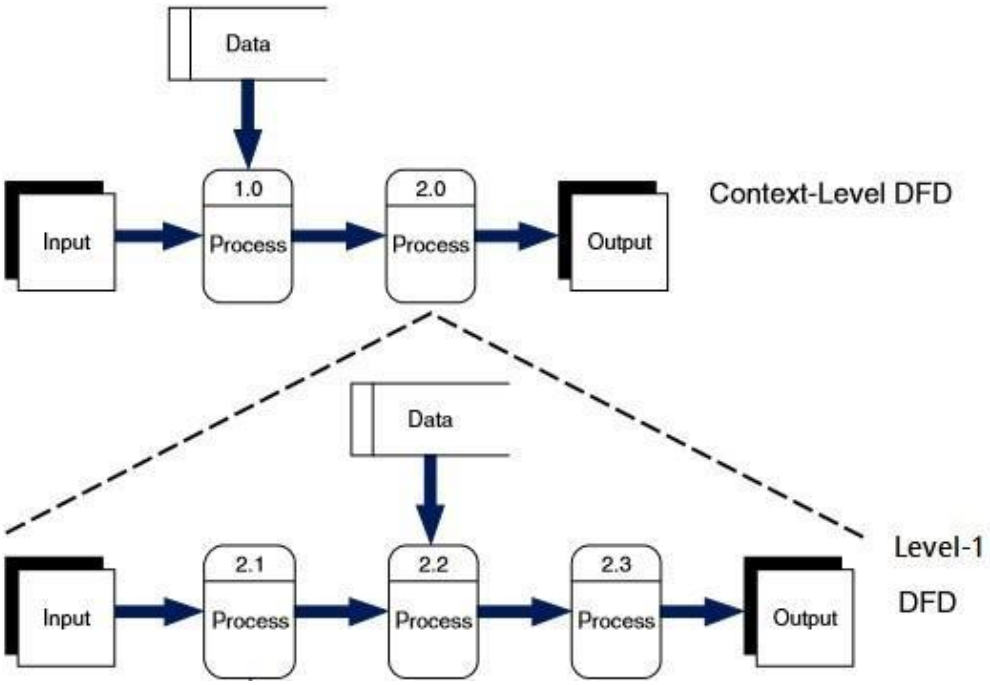
PROJECT DESIGN PHASE-II

DATA FLOW DIAGRAM (DFD)

◈ Level-0 DFD (Context Diagram)



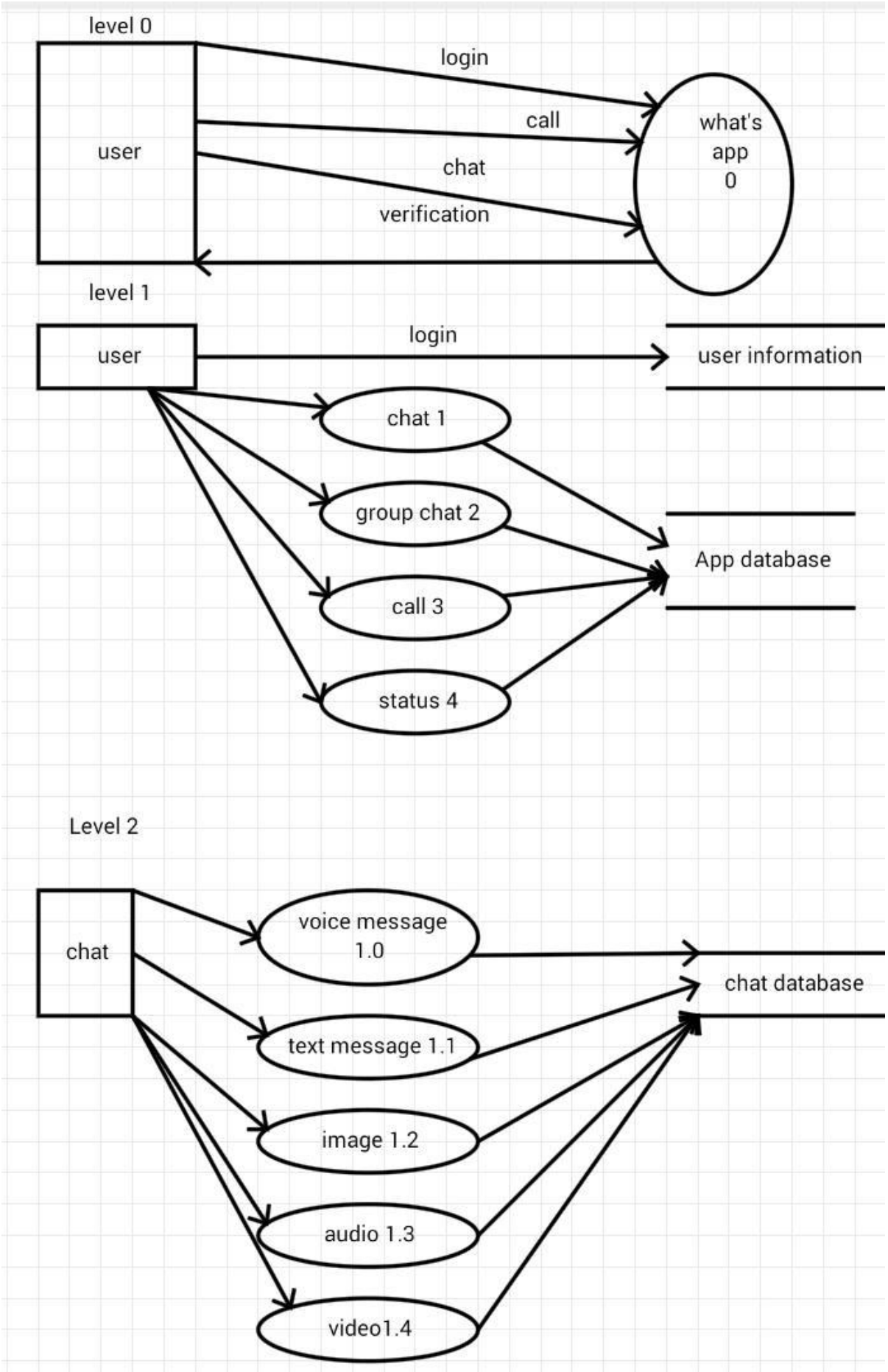




📝 **Explanation:**

- User interacts with system
- System handles:
    - Authentication
    - Video communication
    - Chat
    - Database storage

◈ **Level-1 DFD**


level 0
user
login
call
chat
verification
what's app 0
level 1
user
login
user information
chat 1
group chat 2
call 3
status 4
App database
Level 2
chat
voice message 1.0
text message 1.1
image 1.2
audio 1.3
video1.4
chat database

How to Design a Chat Application?

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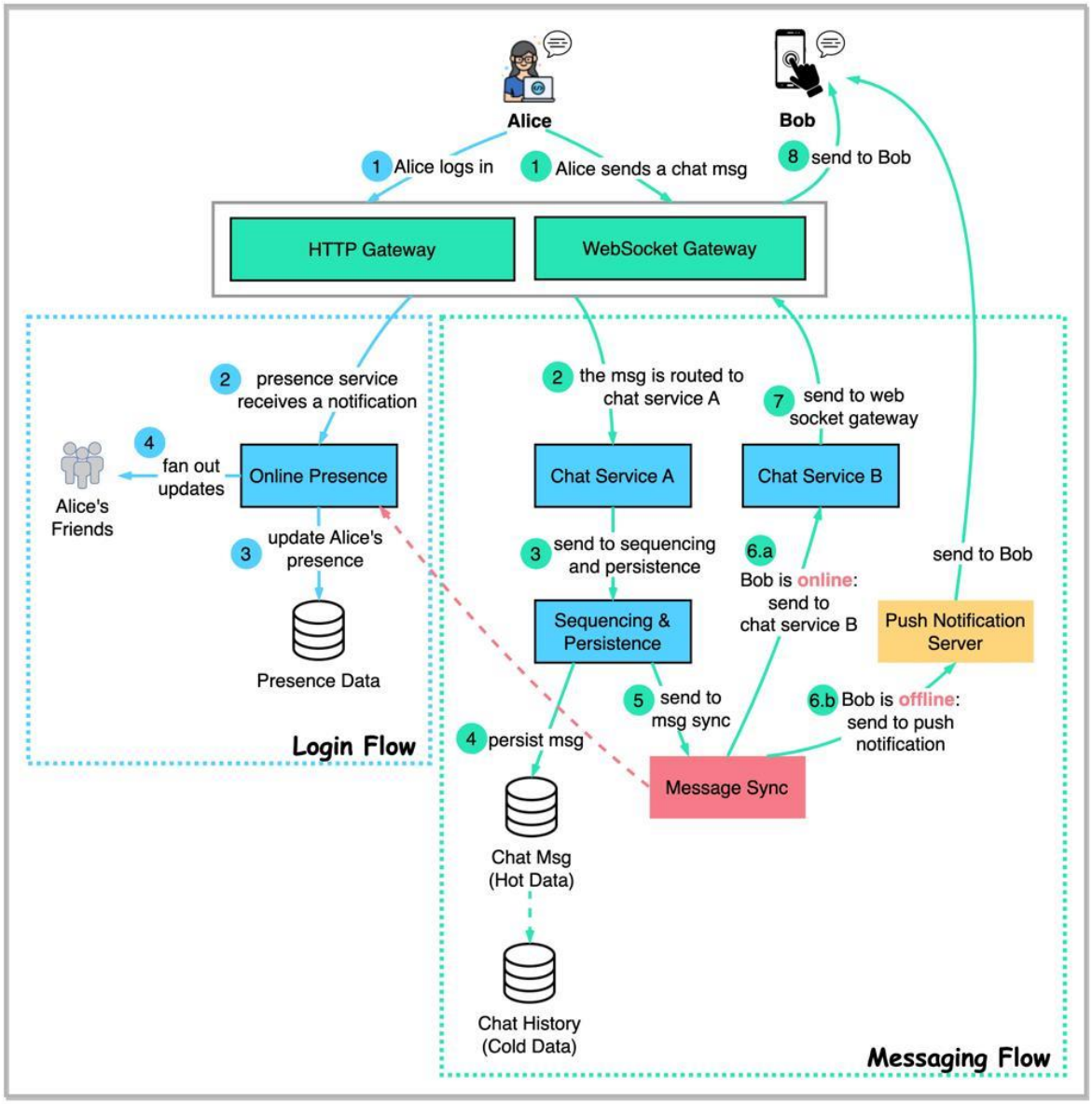
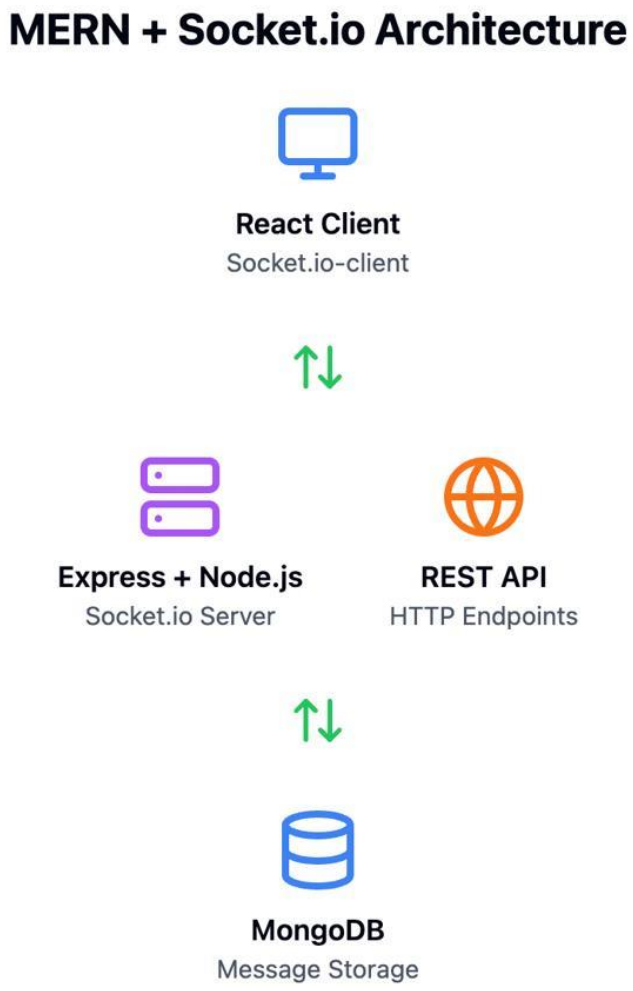




📝 **Processes:**

1. User Authentication
2. Room Management
3. Video Streaming (Agora)
4. Chat System (Socket.io)
5. Database Storage (MongoDB)

📋 **USER STORIES**

Writing

📋 **User Stories**

User Type	Functional Requirement	User Story No	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Web User)	Registration	USN-1	As a user, I can register using email and password	User account created successfully	High	Sprint-1
Customer	Registration	USN-2	As a user, I receive confirmation after registration	Confirmation message displayed	High	Sprint-1
Customer	Login	USN-3	As a user, I can login using credentials	User redirected to dashboard	High	Sprint-1
Customer	Meeting	USN-4	As a user, I can create a meeting room	Room ID generated	High	Sprint-1
Customer	Meeting	USN-5	As a user, I can join a meeting using ID	Successfully enters meeting	High	Sprint-1
Customer	Chat	USN-6	As a user, I can send messages during meeting	Messages visible to all users	High	Sprint-1
Customer	Video	USN-7	As a user, I can enable/disable camera	Camera toggles properly	High	Sprint-1
Customer	Audio	USN-8	As a user, I can mute/unmute microphone	Audio control works	High	Sprint-1

Customer	Screen Share	USN-9	As a user, I can share screen	Screen visible to others	Medium	Sprint-2
Customer	Recording	USN-10	As a user, I can record meeting	Recording saved/downloaded	Low	Sprint-3

📄 FUNCTIONAL REQUIREMENTS

Writing

📄 Functional Requirements

FR No	Functional Requirement	Sub Requirements
FR-1	User Registration	Form-based registration, Gmail login
FR-2	User Authentication	Login with JWT, password encryption
FR-3	Room Management	Create room, join room, leave room
FR-4	Video Communication	Video/audio streaming using Agora
FR-5	Chat System	Real-time messaging using Socket.io
FR-6	Screen Sharing	Share screen in meeting
FR-7	Recording	Record meeting video
FR-8	Profile Management	View/edit user profile

📄 NON-FUNCTIONAL REQUIREMENTS

Writing

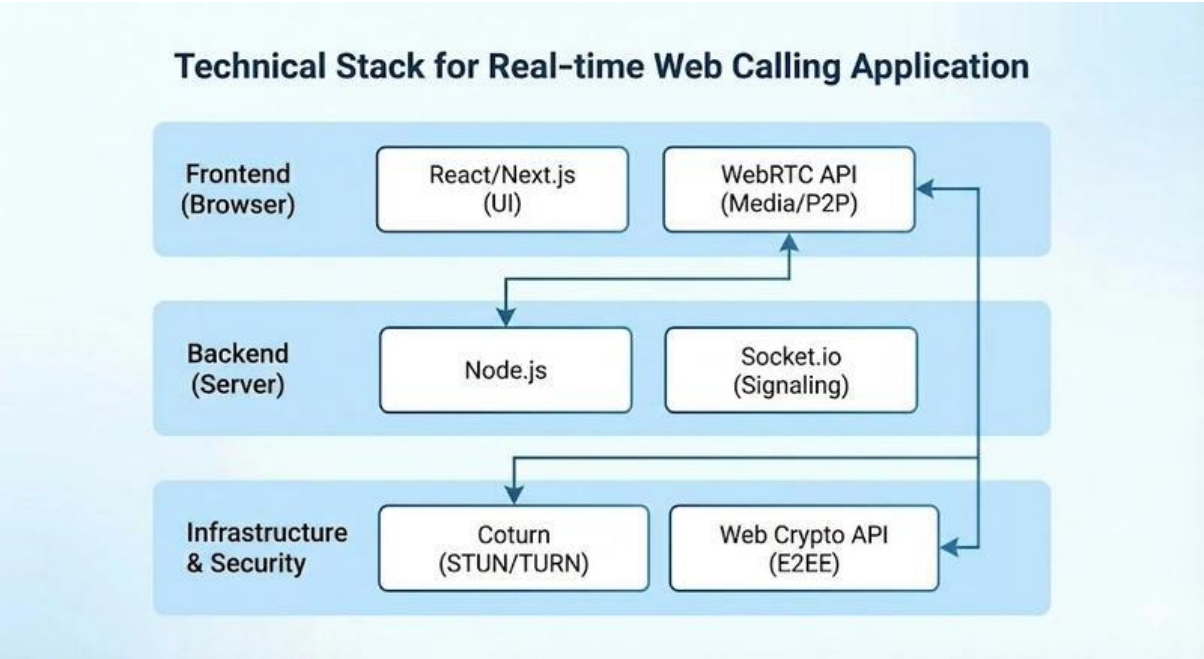
📄 Non-Functional Requirements

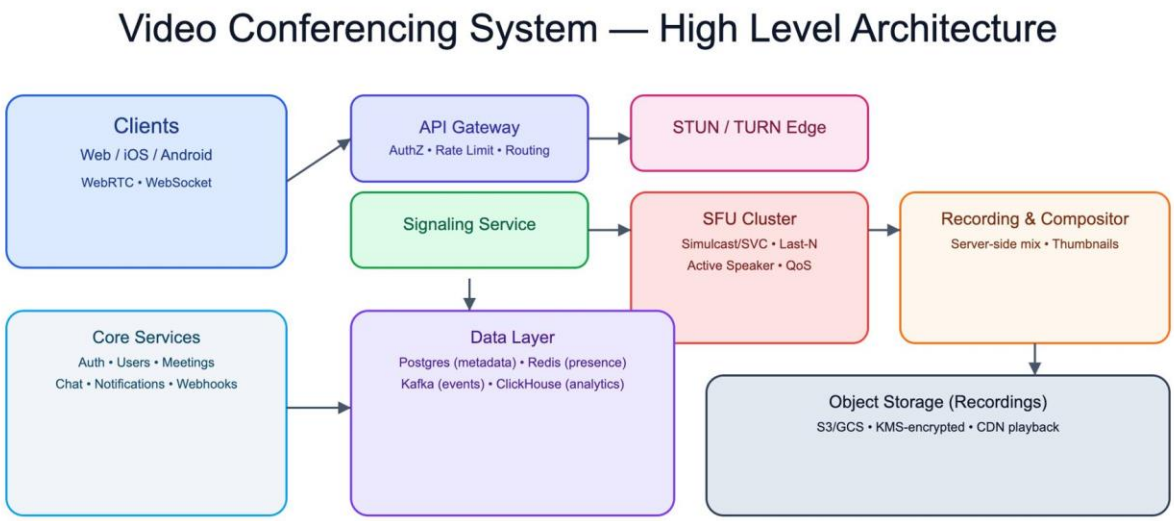
NFR No	Requirement	Description
NFR-1	Usability	Simple and user-friendly interface
NFR-2	Security	JWT authentication, encrypted passwords
NFR-3	Reliability	Stable video and chat communication
NFR-4	Performance	Low latency video/audio streaming
NFR-5	Availability	System available 24/7

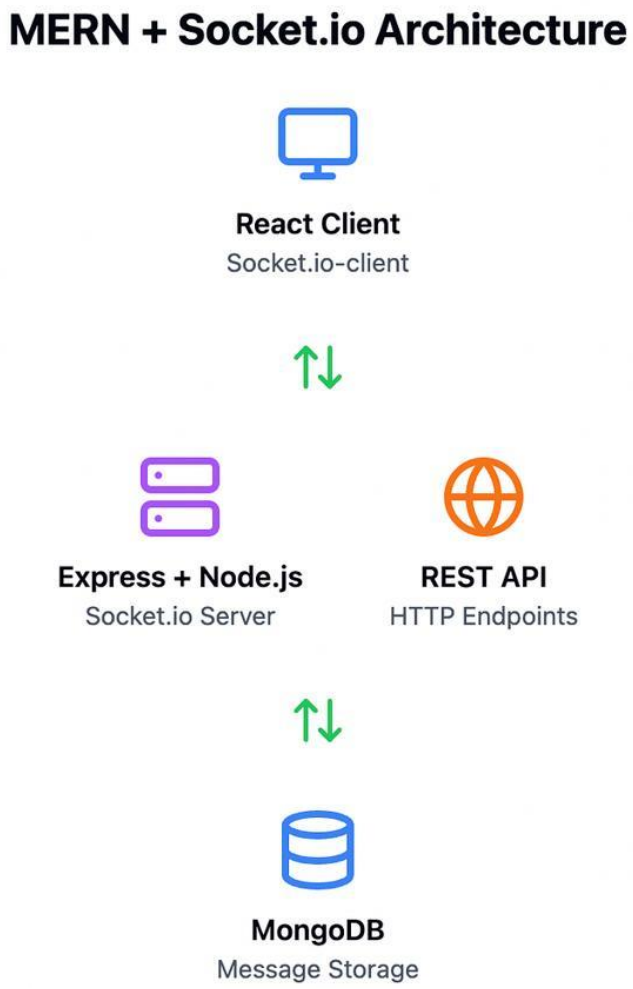
NFR-6	Scalability	Supports multiple users and rooms

🏗 TECHNICAL ARCHITECTURE

🔷 Architecture Diagram







📊 TABLE-1: COMPONENTS & TECHNOLOGIES

Writing

📊 Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web interface for users	React.js, HTML, CSS
2	Application Logic	Backend APIs & logic	Node.js, Express.js
3	Real-time Communication	Chat & events	Socket.io
4	Video Streaming	Audio/video calls	Agora RTC
5	Database	Store user & room data	MongoDB
6	Authentication	Secure login system	JWT, bcrypt

7	API Integration	External services	Agora API
8	Infrastructure	Hosting & deployment	Localhost / Cloud

📊 TABLE-2: APPLICATION CHARACTERISTICS

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📊 Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open Source Frameworks	Used frameworks	React, Node.js
2	Security	Data protection	JWT, Encryption
3	Scalable Architecture	Multi-user support	MERN Stack
4	Availability	24/7 uptime	Cloud/Server
5	Performance	Fast response	Socket.io, Agora